

D868

Analytical Chemistry with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A certified reference tablet contains 50.0 mg of caffeine. A contract lab with ISO accreditation and a ten-year client list reports four replicates: 52.4, 52.5, 52.4, and 52.6 mg. How are these results best described?

- A. Accurate but not precise
- B. Neither accurate nor precise
- C. Precise but not accurate
- D. Accurate and precise

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

The true value is 50.0 mg. Which set of four results is accurate but not precise?

- A. 44.0, 45.1, 46.2, 43.9
- B. 52.4, 52.5, 52.4, 52.6
- C. 46.0, 54.1, 49.8, 50.1
- D. 49.9, 50.0, 50.1, 50.0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A dye solution looks purple and is measured at 525 nm on a spectrophotometer purchased in 2019. In a 1.00 cm cuvette its absorbance is 0.360. The dye's molar absorptivity at that wavelength is $12,000 \text{ M}^{-1}\text{cm}^{-1}$. What is its concentration?

- A. $6.86 \times 10^{-4} \text{ M}$
- B. $3.00 \times 10^{-5} \text{ M}$
- C. 0.360 M
- D. $4.32 \times 10^3 \text{ M}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A sample of 1.50×10^{-5} M with $\epsilon = 15,000 \text{ M}^{-1}\text{cm}^{-1}$ reads $A = 0.450$. What path length was used?

- A. 2.00 cm
- B. 3.00×10^{-5} cm
- C. 0.500 cm
- D. 0.450 cm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Standards prepared by serial dilution from a 1000 ppm stock give the calibration line $A = 0.0500c + 0.010$, where c is in ppm, with $R^2 = 0.998$. An unknown reads $A = 0.410$. What is its concentration?

- A. It cannot be reported, because R^2 is below 1.000.
- B. 0.0305 ppm
- C. 8.00 ppm
- D. 8.20 ppm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A 25.00 mL sample of sulfuric acid, H_2SO_4 , is titrated with 0.1000 M NaOH using phenolphthalein, which turns faint pink at 30.00 mL. The burette was rinsed twice with the titrant. What is the acid's concentration? ($\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$)

- A. 0.06000 M
- B. 0.1200 M
- C. 0.1000 M, the same as the base at equivalence
- D. 0.08333 M

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

What volume of 0.200 M HCl is needed to neutralize 40.0 mL of 0.150 M $\text{Ba}(\text{OH})_2$?

- A. 30.0 mL
- B. 40.0 mL
- C. 53.3 mL
- D. 60.0 mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A TLC plate 10.0 cm tall is developed for 15 minutes. The solvent front travels 8.0 cm from the origin line and a spot travels 3.6 cm. What is the spot's R_f ?

- A. 2.2
- B. 0.36
- C. 4.4 cm
- D. 0.45

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

On a silica TLC plate, compound P has $R_f = 0.20$ and compound Q has $R_f = 0.75$. Which compound is more polar?

- A. P
- B. They are equally polar, since they ran in the same solvent.
- C. It cannot be judged without their molar masses.
- D. Q

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An analytical balance bought used from a university surplus sale has a readability of 0.001 g. Checked against a certified 10.000 g mass, it reads 10.150 g, and every other weighing that day also runs 0.150 g high. What is the best fix?

- A. Report the masses with more decimal places.
- B. Recalibrate the balance against the certified mass.
- C. Weigh each sample more times and average.
- D. Nothing; every measurement always has some error.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A galvanic cell has a zinc strip in Zn^{2+} solution and a copper strip in Cu^{2+} solution, joined by a wire and a salt bridge soaked in potassium nitrate. The voltmeter reads 1.10 V. After several hours, which electrode has lost mass?

- A.** The copper electrode
- B.** Neither, because electrons only move through the wire
- C.** Both lose equal mass
- D.** The zinc electrode

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An analyst measured absorbance at six known nitrate concentrations to build a calibration for a report due to a state regulator. The report will be printed in color. Which display is most appropriate?

- A.** A bar graph with one bar per concentration
- B.** A scatter plot of absorbance vs. concentration with a best-fit line
- C.** A table only, because graphs always hide precision
- D.** A pie chart of the absorbance values

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A certified reference tablet contains 50.0 mg of caffeine. A contract lab ~~with ISO accreditation and a ten-year client list~~ reports four replicates: 52.4, 52.5, 52.4, and 52.6 mg. How are these results best described?

Target word: accuracy and precision of the data set **Decides:** Results agree closely with each other but all sit about 2.5 mg above the true value.

A	Inverted operation	Reverses the two terms.
B	Adjacent category	The replicates agree closely, so they are precise.
C	Correct	Tight spread (0.2 mg) but a mean of 52.5 mg, well off 50.0 mg.
D	Camouflage	Trusts the lab's credentials instead of comparing to the true value.

2. Answer C Concept 1 • transfer

The true value is 50.0 mg. Which set of four results is accurate but not precise?

Target word: pick the data set that fits the label **Decides:** Mean near 50.0 with a wide spread.

A	Confusion set	Mean about 44.8 mg; neither accurate nor very precise.
B	Inverted operation	Precise but not accurate.
C	Correct	Mean = $200.0 / 4 = 50.0$ mg, but the spread is about 8 mg.
D	Adjacent category	Both accurate and precise.

3. Answer B Concept 2

A dye solution ~~looks purple and~~ is measured at 525 nm ~~on a spectrophotometer purchased in 2019~~. In a 1.00 cm cuvette its absorbance is 0.360. The dye's molar absorptivity at that wavelength is $12,000 \text{ M}^{-1}\text{cm}^{-1}$. What is its concentration?

Target word: concentration **Decides:** $c = A / (\epsilon b)$.

A	Camouflage	Divides 0.360 by the 525 nm wavelength, which is not a term in $A = \epsilon bc$.
B	Correct	$0.360 / (12,000 \times 1.00) = 3.00 \times 10^{-5} \text{ M}$.
C	Wrong stem number	Reports the absorbance as the concentration.
D	Inverted operation	Multiplies A by ϵ instead of dividing.

4. Answer A Concept 2 • transfer

A sample of $1.50 \times 10^{-5} \text{ M}$ with $\epsilon = 15,000 \text{ M}^{-1}\text{cm}^{-1}$ reads $A = 0.450$. What path length was used?

Target word: solve for path length **Decides:** $b = A / (\epsilon c)$.

A	Correct	$\epsilon c = 0.225$; $0.450 / 0.225 = 2.00 \text{ cm}$.
B	Dropped element	$0.450 / 15,000$ leaves out the concentration.
C	Inverted operation	$0.225 / 0.450$ flips the ratio.
D	Wrong stem number	Reports the absorbance as the path length.

5. Answer C Concept 3

Standards prepared by serial dilution from a 1000 ppm stock give the calibration line $A = 0.0500c + 0.010$, where c is in ppm, with $R^2 = 0.998$. An unknown reads $A = 0.410$. What is its concentration?

Target word: unknown concentration from the line **Decides:** $c = (0.410 - 0.010) / 0.0500$.

A	Absolute language	$R^2 = 0.998$ is an excellent fit; real data are never perfect.
B	Inverted operation	Plugs the absorbance in as c and solves for A .
C	Correct	$0.400 / 0.0500 = 8.00$ ppm.
D	Dropped element	$0.410 / 0.0500$ ignores the intercept.

6. Answer A Concept 4

A 25.00 mL sample of sulfuric acid, H_2SO_4 , is titrated with 0.1000 M NaOH using phenolphthalein, which turns faint pink at 30.00 mL. The burette was rinsed twice with the titrant. What is the acid's concentration? ($H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$)

Target word: analyte molarity **Decides:** 1 mol H_2SO_4 reacts with 2 mol NaOH.

A	Correct	$0.1000 \times 0.03000 = 0.003000$ mol NaOH; $\div 2 = 0.001500$ mol acid; $\div 0.02500$ L = 0.06000 M.
B	Dropped element	Uses a 1:1 ratio and skips dividing by 2.
C	Confusion set	Equivalence means matched moles by ratio, not equal concentrations.
D	Inverted operation	Swaps the volumes ($0.1000 \times 25.00 / 30.00$).

7. Answer D Concept 4 • transfer

What volume of 0.200 M HCl is needed to neutralize 40.0 mL of 0.150 M $Ba(OH)_2$?

Target word: solve for titrant volume **Decides:** Each $Ba(OH)_2$ supplies 2 OH^- .

A	Dropped element	Ignores the second hydroxide.
B	Wrong stem number	Repeats the base volume.
C	Inverted operation	$0.200 \times 40.0 / 0.150$ puts the molarities in the wrong places.
D	Correct	$0.150 \times 0.0400 = 0.00600$ mol $Ba(OH)_2 \rightarrow 0.0120$ mol $OH^- \rightarrow 0.0120 / 0.200 = 0.0600$ L.

8. Answer D Concept 5

A TLC plate 10.0 cm tall is developed for 15 minutes. The solvent front travels 8.0 cm from the origin line and a spot travels 3.6 cm. What is the spot's R_f ?

Target word: R_f value **Decides:** Spot distance over solvent-front distance.

A	Inverted operation	$8.0 / 3.6$ flips the ratio; R_f is never above 1.
B	Camouflage	Divides by the plate height, not the solvent front.
C	Adjacent category	The gap between spot and front is a distance, not a ratio.
D	Correct	$3.6 / 8.0 = 0.45$.

9. Answer A Concept 5 • transfer

On a silica TLC plate, compound P has $R_f = 0.20$ and compound Q has $R_f = 0.75$. Which compound is more polar?

Target word: polarity from R_f **Decides:** Polar compounds stick to polar silica and travel less.

A	Correct	The lower R_f means stronger attraction to the polar stationary phase.
B	Absolute language	Same solvent does not mean same polarity; different R_f shows a difference.
C	Adjacent category	TLC separation on silica is governed mainly by polarity.
D	Inverted operation	Farther travel means weaker attraction to silica, so less polar.

10. Answer B Concept 6

An analytical balance ~~bought used from a university surplus sale has a readability of 0.001 g.~~ Checked against a certified 10.000 g mass, it reads 10.150 g, and every other weighing that day also runs 0.150 g high. What is the best fix?

Target word: remedy for the error type **Decides:** A constant offset in one direction is systematic error.

A	Adjacent category	More digits do not make a biased reading correct.
B	Correct	Calibration removes a systematic offset.
C	Confusion set	Averaging reduces random error; a constant offset stays in the mean.
D	Absolute language	Systematic error can be found and corrected.

11. Answer D Concept 7

A galvanic cell has a zinc strip in Zn^{2+} solution and a copper strip in Cu^{2+} solution, joined by a wire and a salt bridge ~~soaked in potassium nitrate. The voltmeter reads 1.10 V.~~ After several hours, which electrode has lost mass?

Target word: which electrode loses mass **Decides:** Zinc is oxidized at the anode: $Zn \rightarrow Zn^{2+} + 2e^-$.

A	Inverted operation	Copper is the cathode, where Cu^{2+} is reduced and plated on, so it gains mass.
B	Dropped element	Leaves out the electrode half-reactions that consume and deposit metal.
C	Adjacent category	One electrode is oxidized and the other reduced; they do not change the same way.
D	Correct	Oxidation dissolves zinc atoms into ions at the anode.

12. Answer B Concept 8

An analyst measured absorbance at six known nitrate concentrations to build a calibration for a report ~~due to a state regulator. The report will be printed in color.~~ Which display is most appropriate?

Target word: graph type for continuous paired data **Decides:** Concentration and absorbance are both continuous variables.

A	Adjacent category	Bar graphs suit categories and hide the linear trend.
B	Correct	Shows the relationship and gives the equation used for unknowns.
C	Absolute language	A table can accompany the plot, but graphs do not always hide precision.
D	Confusion set	Pie charts show parts of a whole, which these are not.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Accuracy vs. precision	Accuracy is closeness of the mean to the true value; precision is closeness of repeated results to each other.	Assuming tightly grouped results must be correct.	A known true value and several replicate results.
2	Beer-Lambert law	$A = \epsilon bc$: absorbance is proportional to molar absorptivity, path length, and concentration.	Multiplying when you should divide, or plugging the wavelength into the equation.	Absorbance, ϵ , and a cuvette path length.
3	Using a calibration curve	Solve the best-fit line for concentration: $c = (\text{signal} - \text{intercept}) / \text{slope}$.	Ignoring the intercept, or plugging the signal in as x .	An equation of the form $y = mx + b$ and an unknown's signal.
4	Titration stoichiometry	At the equivalence point, moles of titrant and analyte match the balanced-equation ratio, not equal concentrations.	Using $M_1V_1 = M_2V_2$ when the mole ratio is not 1:1.	A diprotic acid or a base with two OH^- .
5	Chromatography: R_f and polarity	R_f = distance moved by spot / distance moved by solvent front, both from the origin; on polar silica, less polar compounds travel farther.	Dividing by plate height, or thinking polar compounds travel farther on silica.	TLC distances or two R_f values.
6	Systematic vs. random error	Systematic error shifts every result the same direction and is fixed by calibration; random error scatters results and is reduced by averaging.	Trying to average away a constant offset.	"Every reading is high by the same amount."
7	Galvanic cell electrodes	Oxidation happens at the anode, which loses mass; reduction at the cathode, which gains mass; electrons flow anode $\rightarrow$ cathode.	Swapping anode and cathode roles.	Two metals in their ion solutions joined by a wire and salt bridge.
8	Choosing a data display	Two continuous variables go on a scatter plot with a best-fit line; categories go on a bar graph; parts of a whole go on a pie chart.	Using a bar or pie chart for continuous calibration data.	"How should the data be presented?"

Course scope (public description): In "Analytical Chemistry with Lab," students will deepen their understanding of chemistry through the scientific method, data analysis, and data visualization. This course aligns with Utah Competency Areas C2.1.A and C2.1.B, focusing on questioning, methods, argumentation, models, data, and analysis. Students will gain hands-on experience with classical, optical, chromatographic, mass spectrometric, and electrochemical techniques. Prerequisites include a solid foundation in mathematics and prior chemistry courses. Students will critically evaluate experiments, learning to design effective methods and present their data clearly. This course prepares students to gather, process, and communicate chemical data, emphasizing its application in teaching and real-world scenarios. This course helps prepare students in chemistry and related fields.